Weekly Meeting
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μLHC Physics Update

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MG5 Generator-Level Cuts

run_card.dat

```
#*********************************************************************
# BW cutoff (M+/-bwcutoff*Gamma) ! Define on/off-shell for "$" and decay
#*********************************************************************
 15.0  = bwcutoff      ! (M+/-bwcutoff*Gamma)
#*********************************************************************
# Standard Cuts                                                      *
#*********************************************************************
# Minimum and maximum pt's (for max, -1 means no cut)                *
#*********************************************************************
 20.0  = ptj       ! minimum pt for the jets
 10.0  = ptl       ! minimum pt for the charged leptons
 -1.0  = ptjmax    ! maximum pt for the jets
 -1.0  = ptlmax    ! maximum pt for the charged leptons
 {} = pt_min_pdg ! pt cut for other particles (use pdg code). Applied on
 {}     = pt_max_pdg ! pt cut for other particles (syntax e.g. {6: 100,
#
# For display option for energy cut in the partonic center of mass frame
#
#*********************************************************************
# Maximum and minimum absolute rapidity (for max, -1 means no cut)   *
#*********************************************************************
 5.0 = etaj    ! max rap for the jets
 2.5  = etal    ! max rap for the charged leptons
 0.0  = etalmin ! min rap for the charged leptons
 {} = eta_min_pdg ! rap cut for other particles (use pdg code). Applied
 {} = eta_max_pdg ! rap cut for other particles (syntax e.g. {6: 2.5, 23
#*********************************************************************
# Minimum and maximum DeltaR distance                                *
#*********************************************************************
 0.4 = drjl    ! min distance between jet and lepton
 -1.0  = drjlmax ! max distance between jet and lepton
#*********************************************************************
```

General Run Parameters

bwcutoff

The bwcutoff parameter defines the invariant-mass window ($|m - m_0| < \mathrm{bwcutoff} \times \Gamma$) used for resonance treatment.

Selection: A value of **15** is adopted.

Rationale: This wide window is selected to ensure the inclusion of the far Breit-Wigner tail and near-off-shell contributions, which are physically relevant at high collision energies. This setting avoids numerical instabilities in the integrand while capturing interference effects.

$\boldsymbol{\Delta R(j, l)}$

A minimum angular separation is enforced between jets and leptons: $\Delta R = \sqrt{\Delta\eta^2 + \Delta\phi^2}$

Selection: A threshold of $\boldsymbol{\Delta R(j, l) > 0.4}$ is applied.

Rationale: This value aligns with the radius parameter (R=0.4) of the standard anti-k_T jet clustering algorithm used in reconstruction. By enforcing this cut, collinear divergences in the matrix element calculation are eliminated, and the generation of physically indistinguishable parton-lepton pairs is prevented.

Using a ΔR cut without any η cuts can indeed trigger numerical instabilities in MG5, because the integrator then explores extremely forward phase-space regions where kinematic functions can overflow and produce math-domain errors.

Low-p_T Threshold

Object	Practical lower limit today	Projected minimum with HL-LHC upgrades	Feasibility of lower values
Electron	~20–25 GeV	~15–18 GeV	No (fake-rate + trigger limitations)
Muon	~20 GeV	~12–15 GeV	No (multiple scattering + trigger limits)
Jet	~25–30 GeV	~20–25 GeV	No (pile-up dominates)

Challenges at Low Transverse Momentum

The reconstruction of low-p_T particles is complicated by fundamental particle-matter interactions. Key limiting factors include multiple scattering, poor energy/momentum resolution, and high fake rates driven by pile-up. Additionally, low-p_T signals risk being buried under Beam Induced Background (BIB), leading to high trigger rates.

Generator-Level Strategy

- **Motivation**: Although physical reconstruction is challenging below certain thresholds (e.g. ~20-25 GeV for jets), applying strict cuts at the generator level may bias the physics analysis.
- **Selection**: Inclusive thresholds of $\boldsymbol{p_{T,j} > 10}$ **GeV and** $\boldsymbol{p_{T,l} > 5}$ **GeV** are established.
- **Outcome**: These loose constraints facilitate the evaluation of advanced background suppression techniques (such as timing layers) during offline analysis, ensuring that potential low-momentum signatures are not prematurely discarded.

High-η Treshold

Object	Practical upper limit today	Projected maximum	Feasibility of higher values
Lepton	~2.5	~3.0	Very difficult (tracking geometry + $p_T \to 0$)
Jet	~5.0	~6.0-6.5	Very difficult (radiation + shower + ϕ-compression)

Physical Constraints on Acceptance

Extending physical detection coverage to high $|\eta|$ regions presents significant challenges. Particles with high $|\eta|$ are characterized by shallow incident angles ($\theta \to 0$) and vanishing transverse momentum, which eliminates curvature in magnetic fields. Furthermore, these regions suffer from rapidly increasing radiation fluence and distorted shower shapes, limiting sensor survivability and energy resolution.

Generator-Level Strategy

- **Motivation**: Due to the inherent asymmetry of the beam configuration (proton vs. antimuon), the physics processes are significantly boosted towards the proton direction.
- **Selection**: To maximize phase space coverage in the forward region, extended generator-level acceptance cuts of $|\eta_j| < 7$ **and** $|\eta_j| < 4$ are applied.
- **Outcome**: This inclusive approach ensures that all potential forward signals are generated. Geometric restrictions, such as the shielding nozzle, are reserved for the subsequent detector simulation phase rather than being imposed at the hard-process generation level.

Top Production

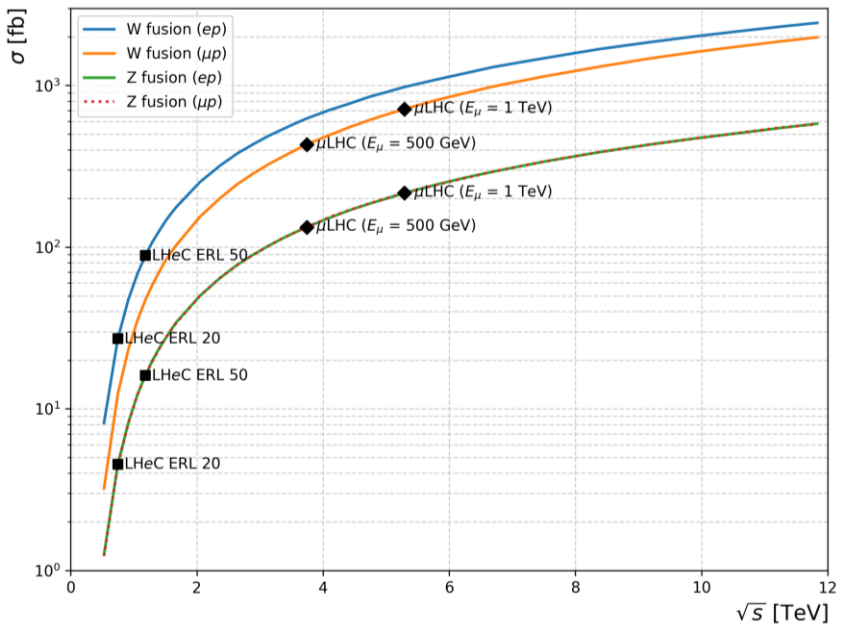
Process	cs [pb] theoretical (no generator-level cut)	cs [pb] flexible cuts	cs [pb] default card
$\mu^+ p \to \bar{\nu}_\mu t X$	30.66	29.26	29.26
$\mu^+ p \to \bar{\nu}_\mu \bar{t} X$	6.2×10^{-7}	6.2×10^{-7}	6.2×10^{-7}
$\mu^+ p \to \mu^+ t X$	2.09	0.84	0.30
$\mu^+ p \to \mu^+ \bar{t} X$	2.09	0.84	0.30
$\mu^+ p \to \mu^+ t \bar{t}$	2.09	0.84	0.30
$\mu^+ p \to \mu^+ t \bar{t} X$	160.23	0.88	0.21
$\mu^+ p \to \bar{\nu}_\mu t \bar{t} X$	8.8×10^{-4}	8.8×10^{-4}	8.6×10^{-4}

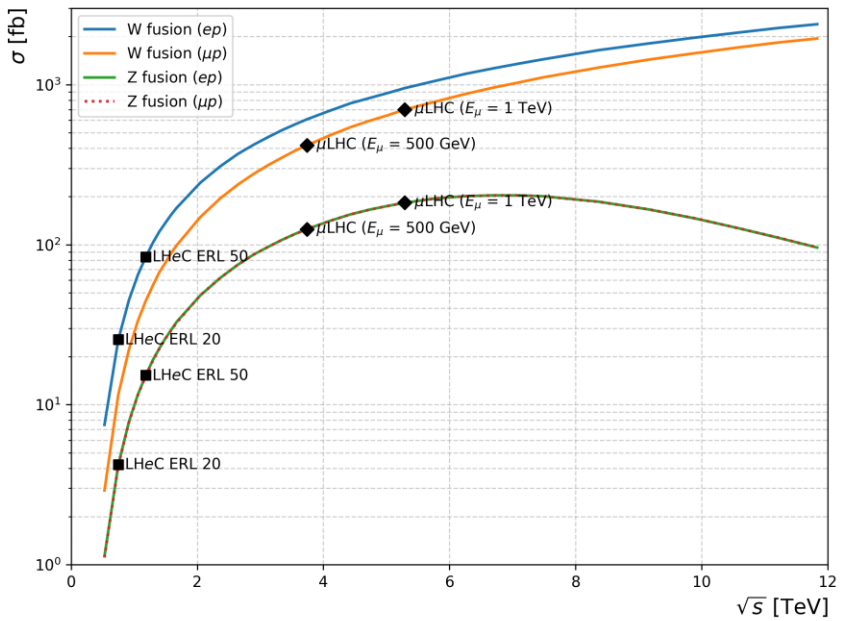
Theoretical
- No generator-level cut

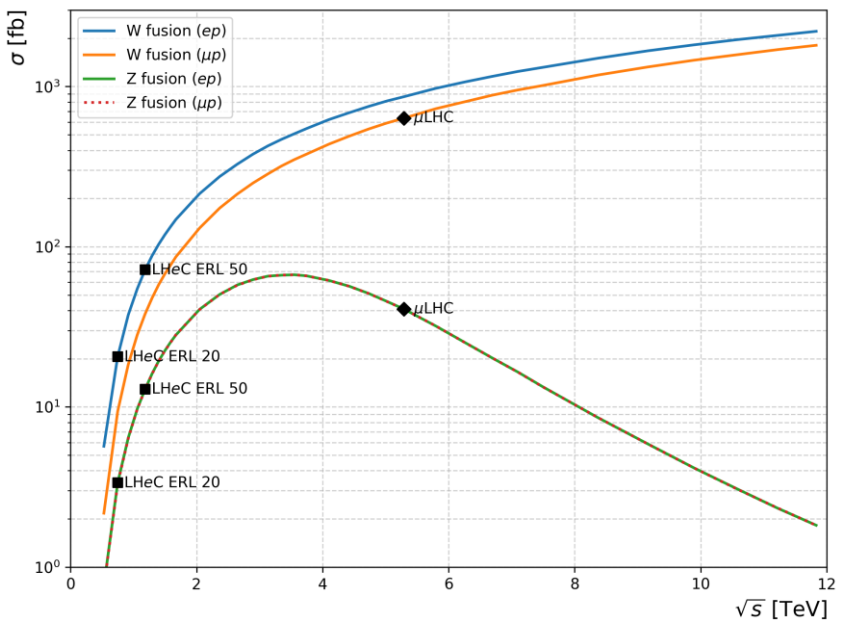
Flexible cuts:
- bwcutoff = 15
- $\Delta R(l,j) = 0.4$
- $|p_{T,j}| > 10$ GeV
- $|\eta_j| > 7$
- $|p_{T,l}| > 5$ GeV
- $|\eta_l| > 4$

Default cuts:
- bwcutoff = 15
- $\Delta R(l,j) = 0.4$
- $|p_{T,j}| > 20$ GeV
- $|\eta_j| > 5$
- $|p_{T,l}| > 10$ GeV
- $|\eta_l| > 2.5$

Higgs Production







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